

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EE/EX-227-CBCS

B.E. IV Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Power System - I

Time : Three Hours

Maximum Marks : 60

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Sketch neat diagram.

1. a) Give the structure and components of power network.
b) What are the various factors involved in power plant economics such as load factor, demand factor, diversity factor, utilization factor?
2. a) What are the various constants used in overhead transmission line?
b) Write about Ferranti effect with the help of phasor diagram.
3. a) Write about Corona formation and explain its advantages and disadvantages.
b) Explain about 'skin effect' and proximity effect in reference to overhead lines.
4. a) An overhead line has the following data:
Span length $\rightarrow$ 160m, Conductor diameter = 0.95cm,
Weight per unit length of the conductor = 0.65 kg/m,
Ultimate stress $\rightarrow$ 4250 kg/cm²,
Wind pressure $\rightarrow$ 40 kg/km² of the projected area, Factor of safety $\rightarrow$ 5. Calculate the sag.

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[2]

- b) What are the types of insulator for overhead transmission line failure of insulator?
- 5. a) Discuss the effect of increase in voltage on the transmission line efficiency.
b) Explain why the voltage drop is of primary concern in the design of the distributors.
- 6. a) Give comparison between isolated and interconnected power system.
b) Prove that transposition of conductors of a three phase transmission line yields equal inductance per phase.
- 7. a) A three phase equilaterally spaced transmission line has a total corona loss of 53 kW at 106kV and a loss of 98kW at 110.9kV. What is the disruptive critical voltage between lines? What is corona loss at 113kV?
b) Draw the phasor diagram of a long transmission line and derive an expression for voltage regulation.
- 8. Write short notes on the following.
 - a) Nominal T and Nominal π representation of transmission line
 - b) String chart and sag template
 - c) Concept of GMR and GMD
